

BICYCLE OPTIMIZATION PROJECT – STAGE 1 – ME303

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Manufacturer: Huffy

Model: Rock Creek 26"

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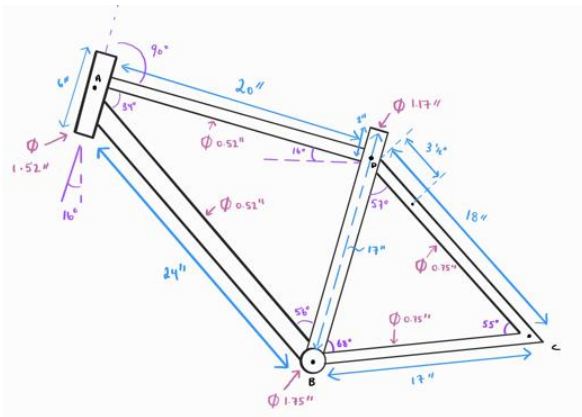
The objective of this study is to analyze the structural performance of a bicycle frame through CAD modeling, static analysis, and stress evaluation to identify potential areas for improvement. A 2D truss model was developed to approximate the frame's structural behavior under static loading conditions.

Dimensional Analysis

The dimensional analysis was completed by measuring the bicycle dimensions and creating a sketch that shows key dimensions for the frame, including all lengths and angles.

- The overall bicycle frame was simplified into simple geometric triangles and angles as shown in figure 1 below.

Dimensional Sketch of the Bicycle Frame



The CAD model of the bicycle frame was created using reference geometry planes at different angles and extending circular profiles with the Boss/Base and Sweep features. Wall thickness was created by offsetting the circular profiles before applying the 3D geometry.

Assumptions

- Chain stay tubing cross-sectional area was simplified to be cylindrical to simplify modelling and subsequent static analysis. The tubing has a very small cross-sectional area; thus, this simplification should have an insignificant impact on the overall stress analysis.

Results

The following figures (3 & 4) provide an isometric and side profile view of the CAD model and a corresponding photo of the bicycle.

Figure 3:

Isometric Profile Comparison of CAD Model and Bicycle Photo



Figure 4:

Side Profile Comparison of CAD Model and Bicycle Photo



Material Analysis

The manufacturer of the bicycle does not explicitly specify the type of steel used for this model. The bicycle itself is an inexpensive, entry level bicycle, so the assumption made is that it will be manufactured with the most common type of steel. According to Recreational Equipment Inc. (REI), carbon steel is commonly used for bike frames (REI, n.d.). Thus, the analysis in this study will be performed under the assumption that the bicycle is manufactured with AISI 1020 carbon steel.

Results

The following table on page 4 provides the mechanical material properties for AISI 1020 carbon steel (cold rolled), per the material science publication Azo Materials.

Table 1.*Material Properties for AISI 1020 Carbon Steel*

Material Property	Value (ksi)
Tensile Strength (σ_t)	60.9
Yield Strength (Y_s)	50.8
Modulus of Elasticity (E)	2.97×10^4
Shear Modulus (G)	1.16×10^4
Poisson's Ratio (ν)	0.29

Note: Poisson's ratio is unitless. (Azom, 2013)

Statics Analysis

Methods

The bicycle frame was modeled as a 2D truss system with pin joints at each connection. A free body diagram (FBD) was created, and the external reaction forces were determined using the equations of equilibrium: summing forces in the x-direction, summing forces in the y-direction, and summing moments about a joint. Once external forces were found, the method of joints was applied to calculate internal forces by isolating each joint, drawing a FBD, applying equilibrium conditions, and solving for unknown forces step by step before moving onto the next joint. Trigonometry was used to resolve forces into horizontal and vertical components to complete the joint analysis.

Assumptions

- Symmetry was assumed for the chain stay tubes connecting to the rear wheel, allowing the 3D bicycle frame to be modeled as a 2D truss system for static analysis. Since both chain stay tubes are identical in length, orientation, and loading conditions, the forces are evenly distributed between them. This assumption simplifies the analysis without significantly affecting accuracy.
- It is assumed the bicycle is stationary, making a static analysis appropriate for resolving the forces acting on the frame. Since no acceleration or motion occurs, dynamic effects are neglected, allowing forces to be analyzed using static equilibrium equations.
- The head tube was removed for the static analysis, and a single joint was formed between the top tube and the downtube. This simplification allows the frame to be analyzed as a statically determinate system. Including the head tube would introduce additional constraints, making the system statically indeterminate, requiring analysis beyond the scope of static equilibrium.
- The rider's weight was applied at the seat tube joint, assuming even distribution. The seat tube is the primary load-bearing location when a cyclist is seated. The static model captures the dominant loading scenario, where the vertical force acts downward on the seat tube, representing the most significant structural loading condition when the bicycle is at rest.

Results

The external reaction forces F_A and F_C of the bicycle frame were calculated to be approximately 96 lb-force at point A and 104 lb-force at point C respectively. The force locations are shown on the free body diagram in figure 5.

The internal forces were calculated as follows: F_{AB} was approximately 298 lb-force in tension, F_{AD} was approximately 256 lb-force in compression, F_{BD} was approximately 1099 lb-force in compression, F_{DC} was approximately 303 lb-force in tension, and F_{BC} was approximately 305 lb-force in tension. A summary of the external and internal reaction forces can be found in figure 6. All hand calculations can be found in Appendix A.

Figure 5.

Freebody Diagram of the Bicycle Frame

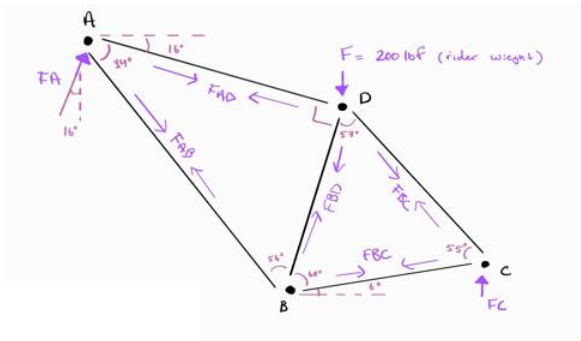


Figure 6.

Spreadsheet Calculations for the Reaction Forces

	A	B	C	D	E	F	G	H
1								
2		Input Values			External Reaction Forces (lbf)		Tension = Positive	
3		Bicyclist Weight (W) [lbf]	200		Reaction at A (F_A)		95.526	
4					Reaction at C (F_C)		104.474	
5								
6		Angles ($^\circ$)			Internal Reaction Forces (lbf)		Compression = Negative	
7		<BAD	34		FAB		296.947	
8		<ABD	56		FAD		-256.101	
9		<DBC	68		FBD		-1098.995	
10		<BCD	55		FDC		302.924	
11		<CDB	57		FBC		304.593	
12		AB from horizontal	34					
13		BC from horizontal	6					
14		AD from horizontal	16					
15		BD from horizontal	74					
16		Headtube	16					
17								
18		Member Lengths (in)						
19		Length AD	20					
20		Length AB	24					
21		Length BC	17					
22		Length BD	15					
23		Length CD	18					
24								
25		Lever Arms						
26		ax	19.897					
27		bx	4.135					
28		cx	16.907					
29		dx	19.225					
30		Length A to C ($ax + cx$)	36.804					
31								
32								

Discussion

The internal force calculations revealed that certain members, such as FBD, experience significantly higher compressive forces compared to others. This result could be expected, as member BD is nearly vertical, meaning it carries a large portion of the downward load, however, it does have an exceptionally high magnitude. This could come from several factors such as an incorrect angle or an algebraic mistake when applying the equilibrium equations. The method of joints relies heavily step-by-step calculations

and one mistake can propagate errors exponentially. Overall, the high tensile and compressive stresses show that additional design considerations could be taken to reinforce the frame to help reduce stress.

Critical Stress Element

Methods

To determine the Critical Stress Element (CSE) at the intersection of the bike crank arm and the chainring, a combined bending and torsion analysis was conducted under a 200lb-force load at the pedal which is in the horizontal position. This analysis was completed using a free body diagram to identify forces and moments acting on the system, moment and stress equations to identify the stresses at the critical element, and a 3D Mohr's circle to determine all three principal stresses and the maximum shear stress.

Assumptions

- The point load is assumed to be completely vertical. This assumption is made for the simplicity of static loading analysis to represent only the force exerted on the pedal by the rider and should impact overall stress analysis as the bicycle is not in motion (no pedaling)..
- The analysis is completed for the right crank arm at the chainring. This allows for analysis of the torsion in a clockwise direction. The direction of the torsion would be opposite (counterclockwise) if analyzing the opposite (left) crankarm/pedal of the bicycle.
- The geometry of the crank arm has been simplified to a simple cylindrical tube configuration from complex cylindrical to square geometry. This allows for simplified calculations for area moment of inertia (I) and polar moment (J). The crank arm is thin and hollow; thus this simplification will have minimal impact on the overall stress analysis.
- Direct shear stress due to the applied force is considered negligible compared to the bending and torsion. The crank arm is a slender beam-like structure with a small cross-section. Due to this, the dominant stresses are from bending moments and torsion rather than from shear force.

Results

To find the critical stress element location at the crank arm and chain ring connection, the shear stress from torsion and the bending stresses are taken into consideration. The singular force acts downward along the y-axis and seven inches from the point of interest on the chainring. This force creates a torsion around the x-axis and a bending moment in the crank arm along the z-axis. Calculations for the bending moment, torsional moment, bending stress and torsion can be found in Appendix A and a summary of these results are shown in Table 2 below.

Table 2.

Calculated Values for Stress Components for the Crankarm and Chainring Interface

Moments	Value (lbf-in)
Bending Moment (Mz)	-600
Torsional Moment (Mx)	1400
Inertia	Value (in ⁴)
Moment Area Inertia ($I_z = I_x$)	0.315
Polar Moment Inertia (J)	0.630
Stress Components	Value (ksi)

Bending Stress (σ_{bend})	3.537
Torsion (T)	-0.758

The clockwise bending moment creates equal tension and compression at the top and bottom points of the neutral axis, and the torsion creates a maximum shear around the outer edge of the circular cross-section. This results in two critical stress locations, one for tension and one for compression. Both locations are shown in figure 7 below, and the tensile stress critical stress element is shown in figure 8.

Figure 7.

Critical Stress Element Locations

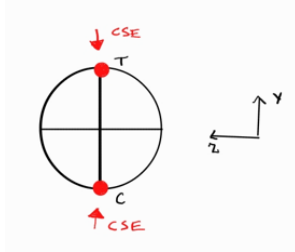
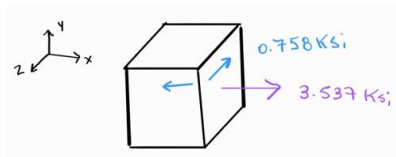


Figure 8.

Tensile Critical Stress Element Diagram



An overview of the stress analysis of the tensile critical stress element is shown in the 3D Mohr's circle diagram in figure 9. Calculations for the principal stresses and maximum shear may also be found in Appendix A at the end of the report. A summary of the stress values including the three principal stresses and the maximum shear stress are listed in Table 3.

Figure 9.

Mohr's Circle for the Tensile Critical Stress Element

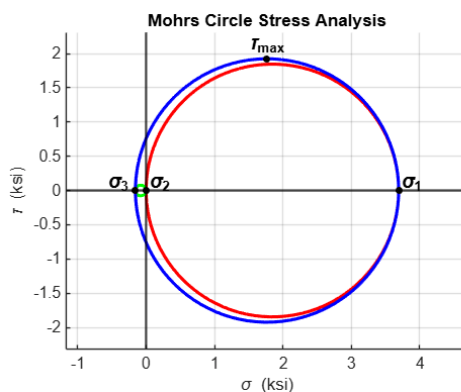


Table 3.*Summarized Results for the Critical Stress Element in Tension*

Stress Component	Value (ksi)
Principle Stress 1 (σ_1)	3.692
Principle Stress 2 (σ_2)	0
Principle Stress 3 (σ_3)	-0.156
Max Shear ($\tau_{\max}$)	1.924

Discussion

The crank arm experiences both torsion and bending, which leads to a combination of stresses that peak at the outermost points of the circular-cross section projected on the interface. There is no additional force that causes increased tension or compression; thus, both are equal in magnitude. This creates two critical stress elements on the interface and highlights that both the top and bottom edges of the crankarm/chainring interface are critical locations for failure and should be considered when designing the system.

Additionally, the vertical force applied to the crank arm resulted in a planar stress configuration, as demonstrated in the 3D Mohr's circle diagram (figure 9). A planar stress state occurs when one of the principal stresses is zero. Recognizing the implications of this zero principal stress in the failure analysis is crucial for designing a system that will not experience premature failure. In the context of this study, identifying the planar stress configuration helps ensure accuracy in stress analysis when designing the crank arm and chainring system. This understanding will aid in the selection of appropriate materials and geometries to minimize stress concentrations and enhance the overall durability and performance of the system.

Deformation

Methods

The methods used to calculate the total vertical deflection of the pedal relative to the crank arm and chainring interface were strain energy using Castigliano's theorem for deflection due to bending and torsion. Additionally, super position was used to find the total deflection. The crank arm system is split into three separate sections. These sections being the pedal spindle where the force acts, the crank arm and the spindle that connects the crank arm to the chainring. Deflection was calculated for each section and then summed to find the overall deflection.

Assumptions

- The geometry of the crank arm has been simplified to a simple cylindrical tube configuration from complex cylindrical to square geometry to aid in simplification of calculations. The crank arm is thin and hollow; thus, this simplification will have minimal impact on the overall stress analysis.
- The torsion is assumed to be constant over the system. The force is applied at a known point and the crank arm system is relatively short, which leads to a constant torque. Thus, we can expect twisting to be uniform across the members that experience torque (AB and BC).

Results

The vertical deflection of the bicycle crank shaft and pedal at the point of the 200 lb-force load was calculated in three-sections as noted in the Methods section. The overall free body diagram of the crank arm system is shown in figure 10. The individual section free body diagrams, along with the moment and deflection calculations can be found in Appendix A.

The deflection in the first section CD (the pedal spindle) was found to be solely from bending which occurred along the x-axis. The deflection in the second section BC (the crank shaft) was from bending along both the x and z axes and torsion around the z-axis. Finally, the deflection in section AB (the spindle that connects the crank shaft to the chainring) occurred from both bending along the x and z axes and torsion around the x-axis. Overall, the total deflection was calculated to be 5.33×10^{-3} inches.

Individual deflection values are listed in Table x below.

Figure 10.

Freebody Diagram of the Bicycle Crank Arm System

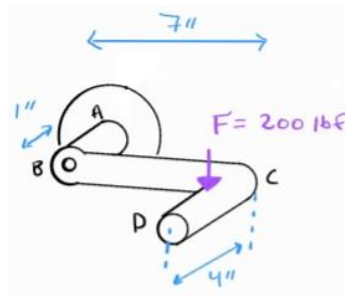


Table 4.

Calculated Values for Deflection in Chain Crank System

Section CD	Value (in)
Deflection from Bending (δ_{CD})	5.70×10^{-5}
Section BC	Value (in)
Deflection from Bending ($\delta_{BC \text{ Bend}}$)	3.04×10^{-3}
Deflection from Torsion ($\delta_{BC \text{ Torsion}}$)	7.66×10^{-4}
Total Deflection (δ_{BC})	3.81×10^{-3}
Section AB	Value (in)
Deflection from Bending ($\delta_{AB \text{ Bend}}$)	1.28×10^{-4}
Deflection from Torsion ($\delta_{AB \text{ Torsion}}$)	1.34×10^{-3}
Total Deflection (δ_{AB})	1.47×10^{-3}
Overall Deflection (δ)	5.33×10^{-3}

Discussion

The total deflection under the 200 lb-force load was small, approximately five-thousandths of an inch (5.33×10^{-3} in), which indicates that the crank arm system is relatively stiff under the given loading

conditions. This stiffness is desirable for maintaining structural integrity in a key component of the bicycle.

A key takeaway from the analysis is the relative impact of bending versus torsional deformations. The largest contribution came from bending in the crank arm itself (member BC). However, torsion in BC and AB played a more significant role than initially expected, with torsional deformation in AB exceeding its bending contribution. This suggests that while bending remains the dominant proponent of deformation, torsion cannot be ignored, particularly in member AB. These insights could inform design improvements, such as increasing the polar moment of inertia (J) in the crank arm spindle (AB) to enhance torsional stiffness or reinforcing the crank arm (BC) to further reduce bending deformation.

Appendix A:

Static Analysis Calculations

Static Equilibrium (External Reactions)

$$\sum F_y = 0 \rightarrow -F + F_A + F_C = 0 \rightarrow F = F_A + F_C \text{ so } 200 \text{ lbf} = F_A + F_C$$

$$\sum M_A = 0 \rightarrow -F(d_x) + F_C(a_x + c_x) = 0$$

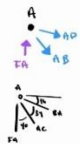
$$F(d_x) = F_C(a_x + c_x)$$

$$F_C = \frac{a_x + c_x}{d_x} = \frac{38.5}{36.809} = 104.472$$

$$F_A = 200 \text{ lbf} - F_C = 200 - 104.472 = 95.528 \text{ lbf}$$

Static Equilibrium (Internal) - Method of Joints

Joint A



$$\sum F_x = 0 \rightarrow F_{AB} \cos 34 + F_{AD} \cos 34 = 0$$

$$F_{AB} = -F_{AD} \left(\frac{\cos 34}{\cos 34} \right)$$

$$\sum F_y = 0 \rightarrow F_A - F_{AB} \sin 34 - F_{AD} \sin 34 = 0$$

$$F_A - \left(-F_{AD} \left(\frac{\cos 34}{\cos 34} \right) \right) \sin 34 - F_{AD} \sin 34 = 0$$

$$F_A + F_{AD} (\cos 34) - F_{AD} \sin 34 = 0$$

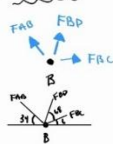
$$F_{AD} = \frac{-F_A}{\cos 34 - \sin 34} = \frac{95.528}{0.375} = 254.10 \text{ lbf}$$

so actually compression

$$F_{AB} = -(-254.10) \left(\frac{\cos 34}{\cos 34} \right)$$

$$F_{AB} = 254.10 \text{ tension}$$

Joint B



$$\sum F_x = 0 \rightarrow -F_{AB} \cos 34 + F_{BD} \cos 34 + F_{BC} \cos 6 = 0$$

$$F_{BD} = \frac{F_{AB} \cos 34 - F_{BC} \cos 6}{\cos 34}$$

$$\sum F_y = 0 \rightarrow F_{AB} \sin 34 + F_{BD} \sin 34 + F_{BC} \sin 6 = 0$$

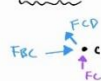
$$F_{BC} \sin 6 = -F_{AB} \sin 34 - F_{BD} \sin 34$$

$$F_{BC} \sin 6 = -F_{AB} \sin 34 - \left(\frac{F_{AB} \cos 34 - F_{BC} \cos 6}{\cos 34} \right) \sin 34$$

$$F_{BC} \sin 6 = -F_{AB} \sin 34 - (F_{AB} \cos 34 - F_{BC} \cos 6) \tan 34$$

$$F_{BC} = \frac{-F_{AB} \sin 34 - (F_{AB} \cos 34) \tan 34}{\sin 6 - \cos 6 \tan 34}$$

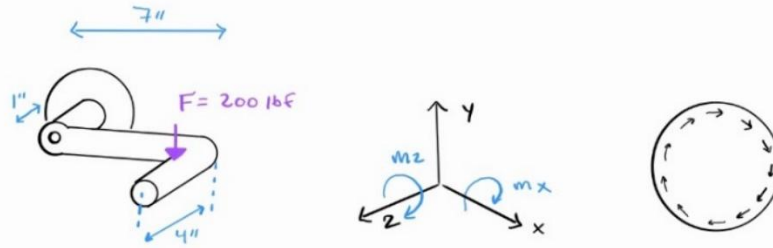
Joint C



$$\sum F_x = 0 \rightarrow F_{BC} \cos 6 - F_{CD} \sin 6 = 0$$

$$F_{CD} = F_{BC} \cos 6$$

Critical Stress Element Calculations



$$\phi = 5/\pi \text{ or } R = 5/2\pi \text{ (all shafts)}$$

$$\text{Thickness } (t) = 3/16 \text{ in}$$

$$I = \frac{\pi}{64} (\text{outer}^4 - \text{inner}^4) = \frac{\pi}{64} \left[(5/\pi)^4 - (5/2\pi - 3/16)^4 \right] = 0.315 \text{ in}^4$$

$$J = \frac{\pi}{32} (\text{outer}^4 - \text{inner}^4) = 2I = 0.630 \text{ in}^4$$

Moments

$$M_x = T = -200 \text{ lbf} (3 \text{ in}) = -600 \text{ lbf-in}$$

$$M_z = (200 \text{ lbf})(7 \text{ in}) = 1400 \text{ lbf-in}$$

Stress components

$$\sigma_{\text{bending}} = \frac{M_x \times C}{I_z} = \frac{1400 \text{ lbf-in} (5/2\pi \text{ in})}{0.315 \text{ in}^4} = 3537 \text{ psi} = 3.537 \text{ Ksi}$$

$$\tau = \frac{T r}{J} = \frac{-600 \text{ lbf-in} (5/2\pi \text{ in})}{0.630} = -758 \text{ psi} = -0.758 \text{ Ksi}$$

Principle stress & Max shear

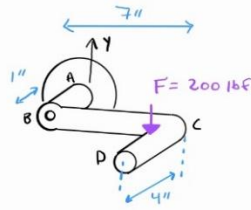
$$\sigma_1, \sigma_2 = \frac{\sigma_x + \sigma_y}{2} \pm \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

$$\sigma_1, \sigma_2 = \frac{3.537 + 0}{2} \pm \sqrt{\left(\frac{3.537 - 0}{2}\right)^2 + (-0.758)^2} \rightarrow \sigma_1 = 3.692 \text{ Ksi}, \sigma_2 = -0.156 \text{ Ksi}$$

$$\begin{aligned} \text{So, } \sigma_1 &= 3.692 \\ \sigma_2 &= 0 \\ \sigma_3 &= -0.156 \end{aligned}$$

$$\tau_{\text{max}} = \frac{\sigma_1 - \sigma_3}{2} = \frac{3.692 - (-0.156)}{2} = 1.924 \text{ Ksi}$$

Deformation Calculations



$$E = 2.97 \times 10^4 \text{ ksi} = 2.97 \times 10^3 \text{ psi}$$

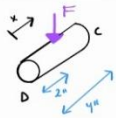
$$I = 0.315 \text{ in}^4 \text{ (from CSE section)}$$

$$J = 0.630 \text{ in}^4$$

$$G = 1.16 \times 10^4 \text{ ksi} = 1.16 \times 10^3 \text{ psi}$$

Find vertical deflection at F

Section C-D



$$\delta_{CD} = \frac{1}{EI} \int_0^L M_{DC} \left(\frac{dM_{DC}}{dF} \right) dx = \frac{1}{EI} \int_0^2 Fx(x) dx = \frac{F}{EI} \int_0^2 x^2 dx$$

$$= \frac{F}{EI} \left(\frac{x^3}{3} \Big|_0^2 \right) = \frac{8F}{3EI}$$

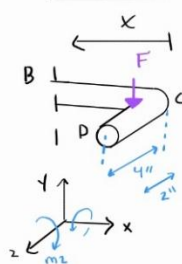
$$= \frac{8(200 \text{ lbf})}{3(2.97 \times 10^3 \text{ psi})(0.315 \text{ in}^4)} = 5.70 \times 10^{-5} \text{ in}$$

Bending

$$M_{DC} = F(x)$$

$$\frac{dM_{DC}}{dF} = x$$

Section B-C



Bending

$$M_x = -F(2)$$

$$\frac{dM_x}{dF} = -(2)$$

$$M_z = F(x)$$

$$\frac{dM_z}{dF} = x$$

$$\delta_{BC} = \frac{1}{EI} \int_0^x M_x \left(\frac{dM_x}{dF} \right) + M_z \left(\frac{dM_z}{dF} \right) dx$$

$$= \frac{1}{EI} \int_0^7 [-F(2)(-2)] + [F(x)(x)] dx$$

$$= \frac{F}{EI} \int_0^7 4 + x^2 dx = \frac{F}{EI} \left(4x + \frac{x^3}{3} \Big|_0^7 \right)$$

$$= \frac{F}{EI} \left(\frac{427}{3} \right) = \frac{427(200 \text{ lbf})}{3(2.97 \times 10^3 \text{ psi})(0.315 \text{ in}^4)}$$

$$= 3.04 \times 10^{-3} \text{ in}$$

Torsion

$$T_{AB} = F \times R = 200 \text{ lbf} (2 \text{ in}) = 400 \text{ lbf-in}$$

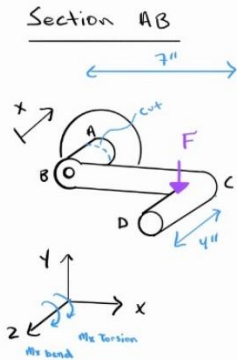
$$\frac{dT_{BC}}{dT_{AB}} = 1 \text{ (T is constant along length of shaft)}$$

$$\theta_{AB} = \frac{1}{GJ} \int_0^7 T \left(\frac{dT_{AB}}{dT_{BC}} \right) dx = \frac{1}{GJ} \int_0^7 T dx = \frac{1}{GJ} \left(400x \Big|_0^7 \right) = \frac{2800}{GJ}$$

$$= \frac{2800 \text{ lbf-in}^2}{(1.16 \times 10^3 \text{ psi})(0.630 \text{ in}^4)} = 3.83 \times 10^{-4} \text{ rad}$$

$$\delta_{BC}^{\text{Torsion}} = \theta_{BC} \times CD/2 = (3.83 \times 10^{-4} \text{ rad})(2 \text{ in}) = 7.66 \times 10^{-4} \text{ in}$$

$$\delta_{BC}^{\text{Total}} = \delta_{BC}^{\text{Bend}} + \delta_{BC}^{\text{Torsion}} = 3.04 \times 10^{-3} + 7.66 \times 10^{-4} = 3.81 \times 10^{-3} \text{ in}$$



Bending

$$M_x = F(2+x)$$

$$\frac{dM_x}{dF} = 2+x$$

$$M_z = F(4x)$$

$$\frac{dM_z}{dF} = 4x$$

$$\begin{aligned} \delta_{AB}^{\text{bend}} &= \frac{1}{EI} \int_0^x M_x \left(\frac{dM_x}{dF} \right) + M_z \left(\frac{dM_z}{dF} \right) dx \\ &= \frac{1}{EI} \int_0^7 [F(2+x)(2+x) + F(4x)(4x)] dx \\ &= \frac{F}{EI} \int_0^7 (2x^2 + 4x + 4) dx \\ &= \frac{F}{EI} \left(\frac{2x^3}{3} + \frac{4x^2}{2} + 4x \right) \bigg|_0^7 \\ &= \frac{F}{EI} \left(\frac{2}{3} + 6 \right) = \frac{18F}{3EI} \\ &= \frac{18(200 \text{ lbf})}{3(2.93 \times 10^3 \text{ psi})(0.315 \text{ in}^4)} \\ &= 1.28 \times 10^{-4} \text{ in} \end{aligned}$$

Torsion

$$T_{AB} = F \times R = 200 \text{ lbf} (7 \text{ in}) = 1400 \text{ lbf-in}$$

$$\frac{dT_{AB}}{dM_{BC}} = 1 \text{ (T is constant along length of shaft)}$$

$$\begin{aligned} \theta_{AB} &= \frac{1}{GJ} \int_0^7 T \left(\frac{dT_{AB}}{dM_{BC}} \right) dx = \frac{1}{GJ} \int_0^7 T dx = \frac{1}{GJ} \left(1400x \right) \bigg|_0^7 = \frac{1400}{GJ} \\ &= \frac{1400 \text{ lbf-in}^2}{(1.16 \times 10^7 \text{ psi})(0.630 \text{ in}^4)} \\ &= 1.92 \times 10^{-4} \text{ rad} \end{aligned}$$

$$\delta_{AB}^{\text{Torsion}} = \theta_{AB} \times BC = (1.92 \times 10^{-4} \text{ rad})(7 \text{ in}) = 1.34 \times 10^{-3} \text{ in}$$

$$\delta_{AB}^{\text{Total}} = \delta_{AB}^{\text{Bend}} + \delta_{AB}^{\text{Torsion}} = 1.28 \times 10^{-4} + 1.34 \times 10^{-3} = 1.47 \times 10^{-3} \text{ in}$$

$$\begin{aligned} \delta_{\text{Total}} &= \delta_{CD} + \delta_{BC}^{\text{Total}} + \delta_{AB}^{\text{Total}} = (5.70 \times 10^{-5} + 3.81 \times 10^{-3} + 1.47 \times 10^{-3}) \text{ in} \\ &= \boxed{5.33 \times 10^{-3} \text{ in}} \end{aligned}$$

References

AISI 1020 carbon steel. (2013). AZoM.com. <https://www.azom.com/article.aspx?ArticleID=9145>.

Recreational Equipment Inc. (n.d.). *Understanding Bike Frame Materials*. REI Co-op. <https://www.rei.com/learn/expert-advice/bike-frame-materials.html>.